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ANNEXURE - II

FORMAT - TECHNICAL SPECIFICATION

For Fully automatic line.

(Capacity – 15,000 liters / day or 25,000 bottles / day @ 600ml / bottle) with all accessories

Sr.No.	Description (Column I)	Required specification (Column II)	Mention your firm Specifications without deviations for all parameters. Mention specifically if any deviations made (Column III)
1.01	***MIXING VESSEL – Working volume 3000 LTR, Gross Volume 3600LTR***		
a	**GENERAL**		
	Material Of Construction	SS 316 contact parts and SS 304 non contact parts.	Yes
	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	Yes
b	**PROCESS DETAILS& OPERATIONAL DATA**		
	Specific Gravity	0.92gms /cc	Yes
	Operating Temperature	Ambient	Yes
c	**DESIGN DATA**		
	Working Volume	**3000 Ltr.**	Yes
	Gross Volume.	**3600 Ltr**	Yes
d	**MATERIAL OF CONSTRUCTION**		
	Shell / Bottom / Top dish	Stainless Steel 316 Grade	Yes

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e	**DRIVE AND AGITATOR DATA**		
	Type of agitator	Baby Anchor with 2 Blade PBT	Yes
	Mixing speed	60 minutes for 3000 liters	Yes
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	Yes
f	**NOZZLES - TOP DISH**		
	Man Hole	450 mm with dished cover and 100 mm sight glass	Yes
	Light Glass	100 MM Pad Type with FLP Vessel lamp	Yes
g	**SHELL**		
	Temperature Sensor	½" BSP	Yes
	Phase Inlet	40 mm with valve	Yes
	Recirculation	50 mm with Valve	Yes
h	**BOTTOM DISH**		
	Product Outlet	50 MM with ball valve	Yes
	Filter	Suitable filter (<200 micron)	Yes
i	**SURFACE FINISH**		
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	Yes
	Product non-contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	Yes

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1.02	***STORAGE VESSEL – Working volume 3000 LTR, Gross Volume 3300LTR***		
	MOC	SS 316 contact parts and SS 304 non contact parts.	Yes
	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	Yes
b	**Process and Operation Data**		
	Gross Volume (L)	3300 Ltrs	Yes
	Working Volume (L)	3000Ltrs	Yes
	Nozzles	Dia450mm, ManHole 50 mmInlet, 40 mm Spare25 mm outlet with valve	Yes
c	**DESIGN DATA**		
	Main Shell ID	ID Dia. 1575 mm	Yes
	Height of Shell / Cone	1500 / 500	Yes
d	**AGITATOR DATA (SUBJECT TO REVALIDATION OF DESIGN DATA)**		
	Type of agitator	Baby Anchor with 2 Blade PBT	Yes
	Agitator Entry	Top Entry	Yes
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	Yes
e	**MATERIAL OF CONSTRUCTION**		
	Product Contact Parts	Stainless Steel 316 Grade	Yes
	Product non-contact parts	Stainless Steel 304 Grade	Yes
f	**BOTTOM DISH**		
	Product Outlet	50 MM with ball valve	Yes
	Filter	Suitable filter (<200 micron)	Yes
g	**SURFACE FINISH**		
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	Yes

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	Product non-contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	Yes
	Storage and Mixing Vessels.	Storage and Mixing vessel should indicate syrup Level.	Yes

2.0	**LIQUID FILLING IN BOTTLE AND INNER PLUGGING AND OUTER CAPPING (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART**		
2.01	***TURN TABLE/BOTTLE FEEDER/CONTAINER LOADER***		
a	**Operation Requirements**		
	Container inputs in turn table by manually and automatically will rotate on disk of turn table and exit through outlet, during rotation of Container will guided by a SS strip, which will guide the container towards outlet path.		Yes
b	**Technical Specifications**		
	Machine type	Fully automatic turn	Yes
	Direction	Clockwise / Anti clock wise as per requirement	Yes
	Power and Motor	0.5 to 1.0 HP/220 to 415 V, 50 to 100 cycles/ second or equivalent	Yes
	Output	100 to 120 Product /Minute (Required speed in filling line, 60BPM)	Yes
c	**Salient Features** All exposed parts are of S.S. 304 to prevent rusting Required conveyor from auto turn table to air-jet cleaning machine to maintain the speed.		Yes

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2.02	**FULLY AUTOMATIC VERTICAL AIR-JET CLEANING MACHINE**		
a	**Technical Specifications**		
	Machine type	Fully Automatic	Yes
	Out put	60 Bottles per Minute	Yes
	Motor Specification	0.5 HP / 415 Volts / 50 Hz. 0.5 HP x 2 = 2 HP / 415 Volts / 50 Hz. or equivalent	Yes
	Cleaning Cycle	Positive air pressure and vacuum cleaning	Yes

2.03	***FULLY AUTOMATIC VOLUMETRIC LIQUID FILLING MACHINE IN BOTTLES***		
	All parts should be made from **SS-316** Materials		Yes
a	**GRAND liquid Bottle size for reference**		
	Bottle size – Height 205mm, Diameter – 235mm, Neck size – 80mm.		Yes
b	**Operation Requirements**		
	Nozzle should go upwards slowly from the bottom level of bottle towards neck during filling to minimize foaming adjustable nozzle is reciprocating according to filling dose.		Yes
c	**Technical Specifications**		
	Machine	Fully automatic	Yes
	Output	Up to 60 Bottles per minute	Yes

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	Filling Volume	600ml	Yes
	Electrical Specification		
	Main Machine	2.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	Yes
	Conveyor	1.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	Yes
	Storage tank	SS316 with the capacity of 100 liters	Yes
	SS conveyor belt made from the SS316 (Length as per the speed of bottles filling – 60BPM) Adjustable Conveyor height as per choice to align with another line machine.		Yes

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2.04	***FULLY AUTOMATIC INNER / OUTER PLUGGING MACHINE***		
	Suitable to capping inner caps output - 60 Bottles/Minute (Bottle outer cap size: Height – above 25mm, Diameter – Above 100mm). One complete set of change parts suitable for any one size diameter of caps & bottle. Total machine construction in SS-304 wherever; Contact parts - SS-316 material.		Yes
a	**Operation Requirements**		
	Outer cap should be placed according to the 60 bottles filling speed.		Yes
b	**Technical Specifications**		
	Machine output	Up to 60 Bottles per minute	Yes
	MOC of machine frame	Mild Steel powder coated and outside cladding SS 304	Yes
	MOC of Contact Parts	Stainless Steel 316	Yes
	Machine Covers	Stainless Steel 304	Yes
	MOC of all vital shafts	EN 31 / Stainless Steel 304/316	Yes
	MOC of Conveyor	Conveyor Slat Stainless Steel 304 Slat with Nylon Strips on Side Conveyor Shafts Mild Steel with Hard Chrome or Zinc Plating Conveyor Sprockets Mild Steel or Nylon	Yes

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a	**Technical Specification**		
	Output	Upto30-60BottlesperMinute	Yes
	No. of Sealing Head	According to the output of 60 BPM	Yes
	Capping Unit	Magnetic Screw Capping Head with Torque Adjustment	Yes
	Caps Feeding	Vibratory Bowl Feeder	Yes
	Caps Hammering	Pneumatic System	Yes
	Conveyor Length	Depending upon the product output of 60BPM	Yes
b	**Machine's material of construction**		
	Machine main frame	Square Pipe, Stainless Steel 304	Yes
	Machine Coverings	Complete in Stainless Steel 304	Yes
	Product Contact Parts	Stainless Steel 316	Yes
c	**Machine's electronics & pneumatics components**		
	Capping Head Motor Make	1 H.P Standard Model	Yes
	Capping Head VFD Make	1 H.P Delta	Yes
d	**Machine's power & air consumption requirement**		
	Power	1 H.P / 230 V/AC or 1.5KW or equivalent	Yes
	Phase	Single Phase/ Three Phase according to the requirement	Yes
	Compressed Air Required	6 bar pressure, 0.5 CFM or equivalent	Yes

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3.0	***LIQUID BOTTLES DRYING TUNNEL (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART*** The fully automatic drying tunnel should be designed with precision, these machines are widely recognized for their durability, design excellence and high performance.		
a	**Technical Specifications**		
	Machine Type	Fully automatic	Yes
	Temp. Req.	100' To 150' Degree Centigrade in Chamber,	Yes
	Heater	Heater With Heating System. Total 10 To 24 Kw	Yes
	Motive Load	1.5 Hp Conveyor and 0.5 Hp Blower Fan Load	Yes
b	**Material**		
	•Incoming MCB 32amp indo copper make or equivalent. •Blower Motor's are remi make or equivalent. •Blower Motor are separately controlled.		Yes

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4.0	***LIQUID BOTTLES PACKING IN CARTON BOX UNIT WITH ACCESSORIES FOR INSTALLATION, PACKING, CLOSING AND STRAPPING FOR TRANSPORTATION – REQUIRED NO. 1 SET***		
	Carton Box size for reference: Length - 470mm, Width – 390mm and Height – 270mm - Yes		
a	***S-Conveyor***		
b	**Technical Specifications**		
	Output per Hour	60 to 65 Bottle/Minute	Yes
	Working Desk Height	860 To 910 mm according to the requirement	Yes
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-11 mtr/min	Yes
c	***Case Erector***		
d	**Technical Specifications**		
	Output per Hour (No. of cases)	8-10/Minute	Yes
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-10 mtr/min	Yes
	Material of construction	MS powder coated	Yes
e	***Roller Conveyor***		
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes

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i	**Idler Roller Conveyor**		
	After case sealing from top of the box filled box will come and accumulate on this gravity roller conveyor. From this point filled box will pick manually		Yes
j	***Carton box strapping machine***		
k	**Technical Specifications**		
	Output per Hour (No. of cases)	2-5/Minute or equal	Yes
	Direction of Movement	Left to Right	Yes
	Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement	Yes
	Power consumption	400 watt	Yes

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	Power consumption	0.37 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes
	Drive roller	SS-304	Yes

f	**Case Tapping Machine**		

g	**Technical Specifications**		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes

h	**Pick and Place Unit**		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	Yes
	Power consumption	0.37 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes

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5.0	***FULLY AUTOMATIC BOTTLE PRINTER TO PRINT PLACE OF MANUFACTURE, DATE OF MANUFACTURE, BATCH CODE AND EXPIRY DATE***		
a	**Technical specifications**		
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	Yes
	Voltage	90 V to 260 V,47 Hz to 63 Hz	Yes
	Resolution	8 dots per mm	Yes
	Print area	30mm to 50mm	Yes
	Message Storage Capability	250	Yes
	Maximum Lines of Print	5, 32pixel	Yes
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	Yes

Ducting:

All Electrical cable of Bottle filling line is proper route and kept in hinge type duct.

There is separate duct for Electrical cable and Pneumatic Pipe.

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